Queue-level capture

What a passive k-lot quoter actually keeps on the favourite end, once queue position and sweep timing are real

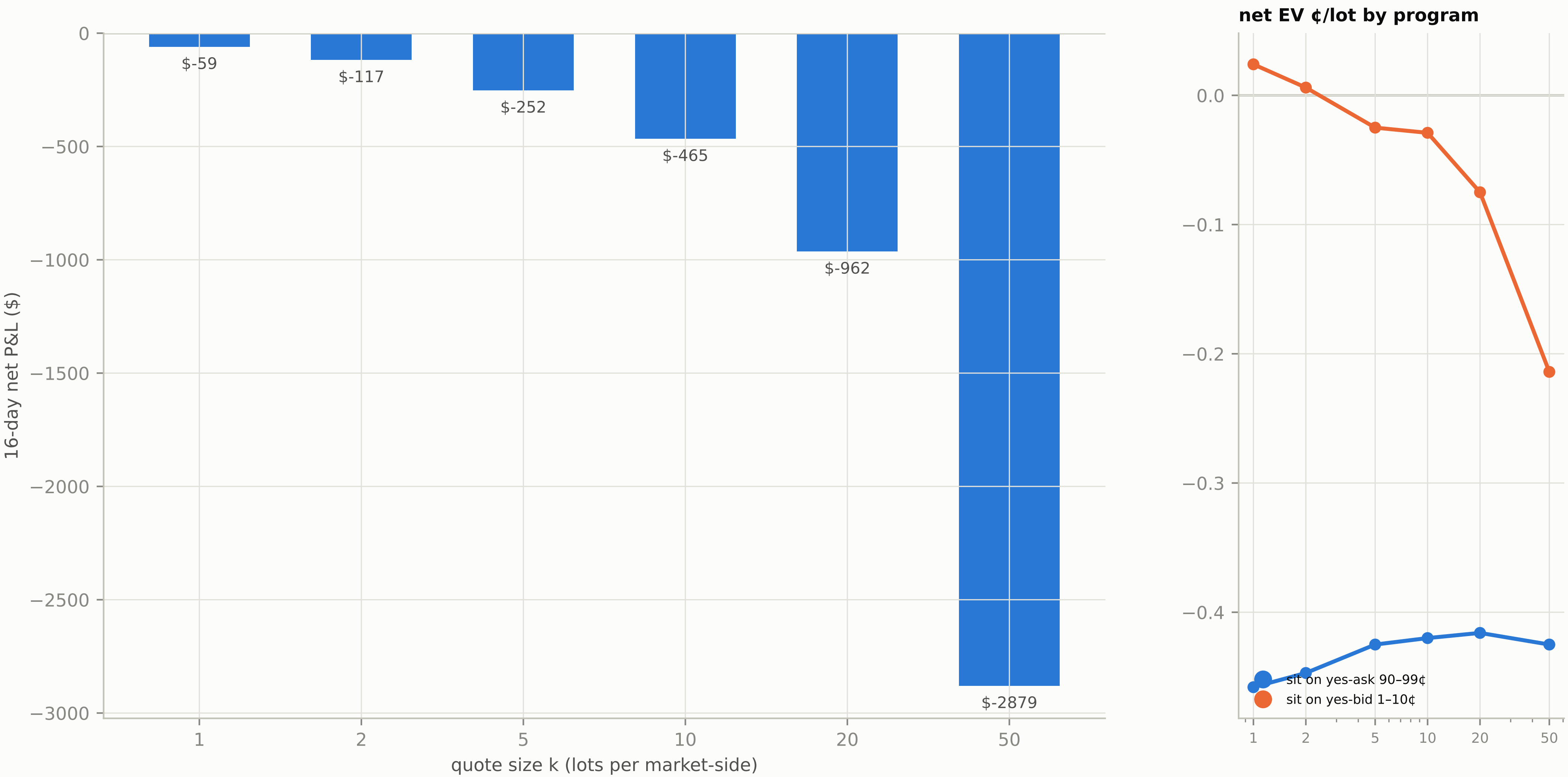
Event-level replay of L1 quotes + prints for 39 fee-netted target books over the 16 sealed days. We join the back of the displayed queue, lose our spot on every level move, get filled only by volume that chews through the queue ahead, and hold to settlement. Maker fees netted per fees.py. Verdict: DOES NOT SURVIVE queueing.

−$465	-0.22¢	208k	−$572
net P&L, k=10 lots, 16 days	net EV per filled lot at k=10	lots filled at k=10 across 25k markets	max drawdown of daily curve, k=10

Context: the zero-queue unit-lot replay (1 lot in every market) made $5,340 over the same days. The number above is what queueing leaves of it at k=10 — the honest capture, not the arithmetic ceiling. Fill events: 640k; markets truncated by the event cap: 0.

C1 · Capture curve — P&L and EV per lot vs quote size k

Bars: 16-day net P&L by k (both programs). Dots: net EV per filled lot. The gap between k and k×(unit EV) is the queue tax.

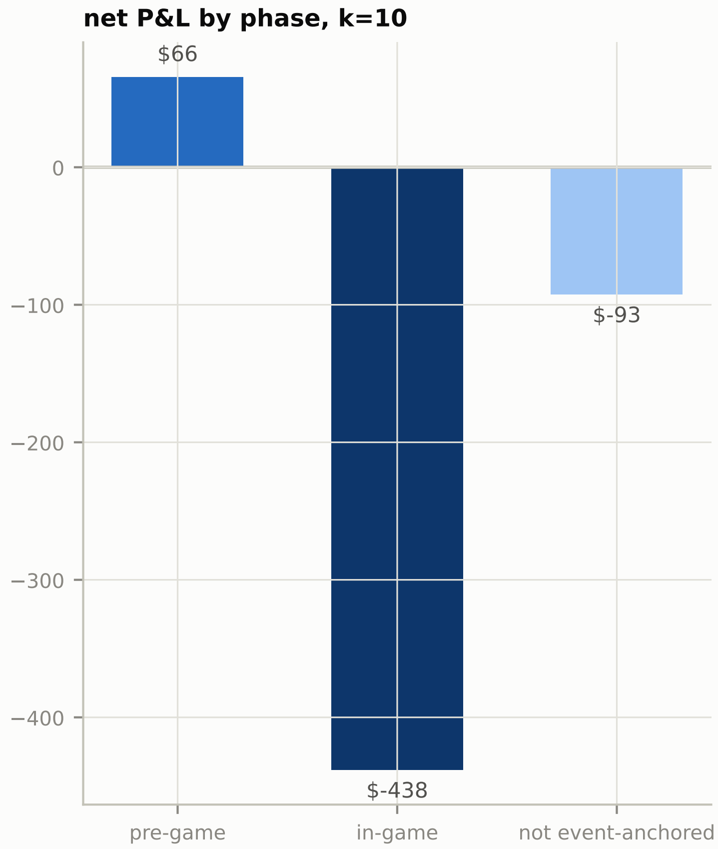


Left: one bar per k, both programs pooled, maker fees netted. Right: EV per filled lot— if EV holds while k grows, size costs volume share, not edge quality; if EV decays with k, later fills are adversely selected. No market impact from our own quote is modelled (replay optimism).

C2 · Four-line worst bill by size, and where fills happen

Loss-budget doctrine numbers for the simulated line at each k; phase split at k=10.

k	total	lots	markets	worst market	worst day	max DD
1	−$59	24k	25k	−$0	−$25	−$68
2	−$117	48k	25k	−$0	−$48	−$134
5	−$252	109k	25k	−$1	−$100	−$305
10	−$465	208k	25k	−$1	−$190	−$572
20	−$962	395k	25k	−$2	−$402	−$1,165
50	−$2,879	904k	25k	−$5	−$1,095	−$3,372

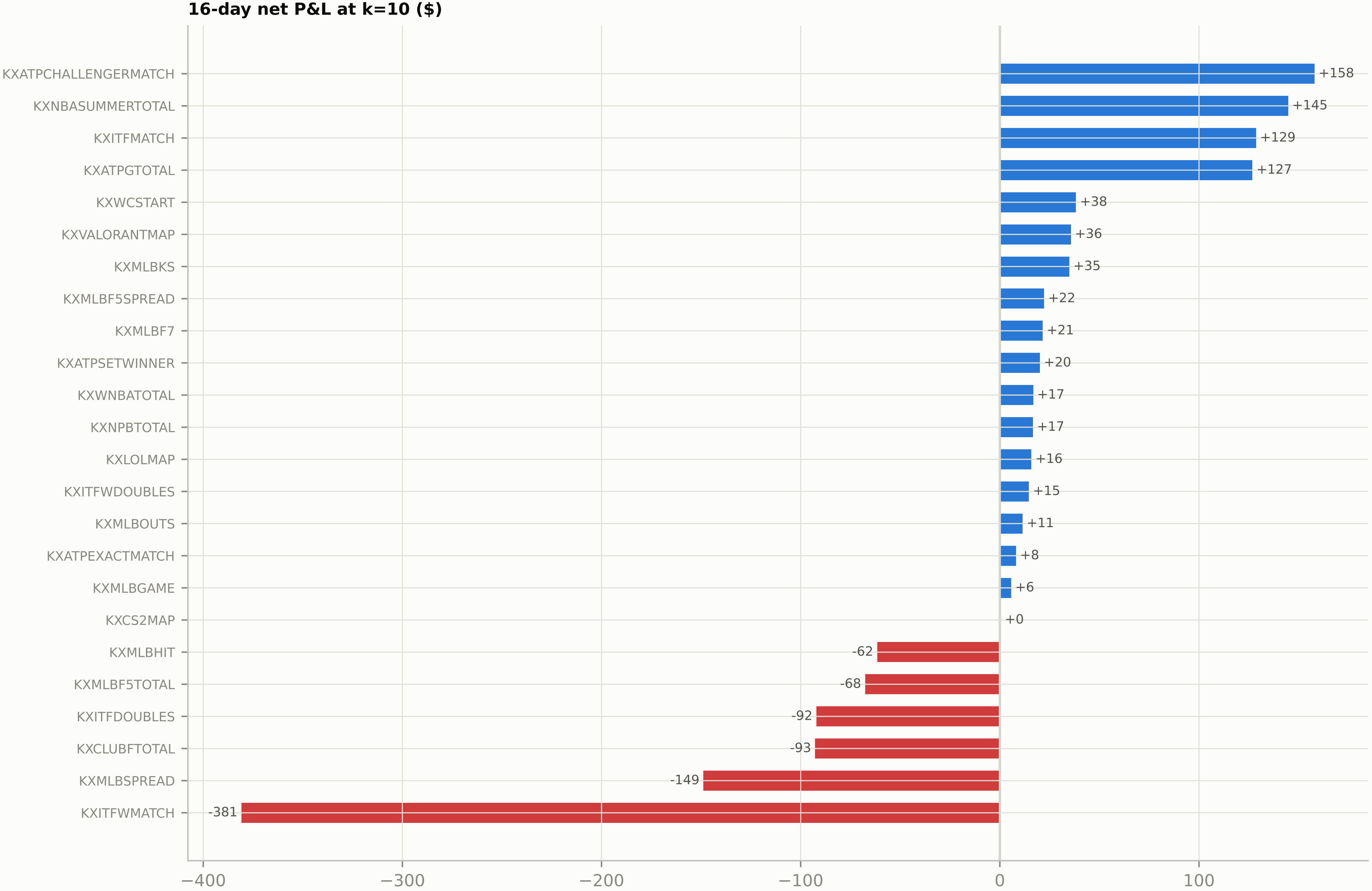


Worst market = biggest per-market loss (both programs, all fills in that market). Daily curve anchored on settlement date. The worst single FILL remains capped at −10¢/lot by construction of the favourite-end band.

C3 · Where the captured money lives — per book at k=10

queue_sim

Top 18 and bottom 6 books by simulated net P&L. Books that die under queueing leave the target list.



Fee-netted. Books with negative simulated capture are removed from the deployment list even if their zero-queue edge was positive— queueing reorders the ranking because thin-queue books fill better.

Method & caveats

Queue model

Join the back of the displayed best-level queue on our side whenever the price is in band (ask program: yes-ask 90–99¢; bid program: yes-bid 1–10¢). A level move re-joins at the new level (queue position lost). Displayed-size drops without a trade clamp queue-ahead to displayed (cancels assumed behind us otherwise). A print at our level first consumes the queue ahead; overflow fills us, up to k. One round per market-side; positions held to settlement; maker fees per fees.py.

What this inherits from the chain

Universe = 39 books, fee-netted positive in the unit-lot replay, minus exclusions (KXMLBRBI, KXMLBSB, KXMLBTB). Settlement truth = catalog_normal 07-24 snapshot. Sealed days 07-10..25 only.

Known optimism

Our quote adds no size to the book and never moves the market; competitors do not react to us; queue-ahead clamping is conservative but re-join-at-back is not perfectly so. Prices are exchange prints — no latency model.

Known pessimism

We never improve the price (always join, never lead), never re-quote after the k-lot round completes, and abandon queue position on every tick of the best level. A real engine does better on all three.

Next gates

Shadow quoting (2 weeks, existing engine shadow rig, target-book universe) must reproduce this capture curve; then the four-line bill at chosen k goes for approval; then probe; then $200 micro-live. This document is a measurement, not an authorisation.